7.A Gases I

Introduction to Gases; Pressure. This chapter focuses on the properties and behavior of gases, the simplest state of matter. We'll start by defining the term "gas" and identifying some empirical variables of interest to gases. Pressure is a particularly important variable used most often in a gaseous context.

- The three common states or phases of matter are **solid**, **liquid**, and **gas**.
- Gas is the *simplest* phase of matter because particles are essentially isolated from each other.
- Gases do not have intrinsic volume; they expand to fill the volume of the container.

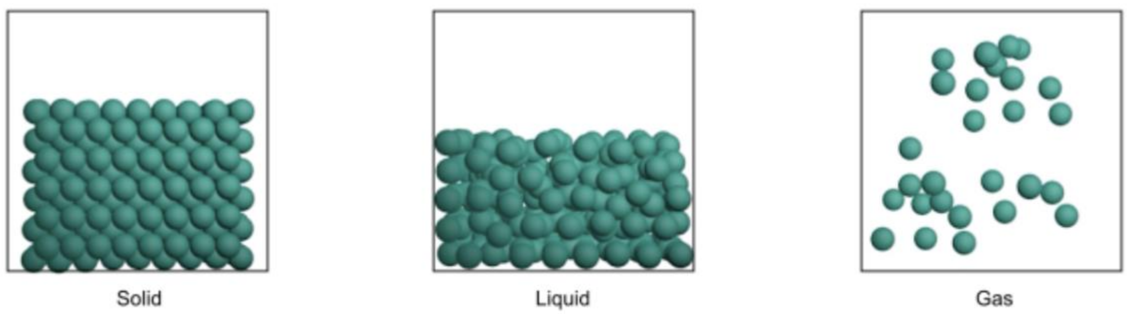


Figure 7.1. Molecular-level pictures of the phases of matter.

- **Pressure** is the force exerted by a gas per unit area over which the force acts.

$$P = F/A = U/V$$

- **Hydrostatic pressure** is pressure exerted by a column of fluid with mass density ρ. Pressure can be expressed in "height of fluid" units.

$$P = \rho \cdot g \cdot h$$

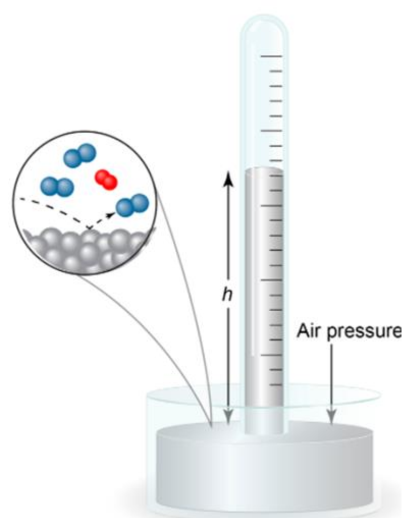


Figure 7.2. A simple barometer.

- Units of pressure include "force per area" units and "height of fluid" units.
    - 1 Pascal (P)
    - 1 millimeter of mercury (mmHg)
    - 1 atmosphere (atm)
    - 1 bar
    - 1 pound per square inch (psi)

- **Standard temperature and pressure (STP)** is a specific set of temperature and pressure conditions: 273.15 Kelvin (0.00 °C) temperature and 1.00 atm pressure.

The Empirical Gas Laws. Observed relations between gas pressure *P*, volume *V*, number of moles *n*, and temperature *T* formed the basis for the development of a set of natural laws governing these variables. These laws are known as the empirical gas laws. We'll soon put them together into a single unifying equation.

- We are interested in relations between four key variables for gases: pressure (*P*), volume (*V*), number of moles (*n*), and temperature (*T*).

- **Boyle's law:** the pressure and volume of a gas are inversely proportional at constant number of moles and temperature.

Functional form: $PV = k$

Linear form: $P = k\left(\frac{1}{V}\right)$

Multi-point form: $P_1V_1 = P_2V_2$

- **Charles's law:** the volume and temperature of a gas are directly proportional at constant number of moles and pressure.

  Functional and linear form: $V = kT$

  Multi-point form: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

- Charles's work also led to the development of the **Kelvin** temperature scale. Kelvin should *always* be used when working problems involving gases.

- The **combined gas law** incorporates the previous two laws and holds true for a gas at constant number of moles. The ratio of *PV* and *T* is constant at constant *n*.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

7.1. The pressure exerted on a sample of a fixed amount of gas is doubled at constant temperature, and then the temperature of the gas in Kelvin is doubled at constant pressure. What is the final volume of the gas?

A. Twice the initial volume
B. Four times the initial volume
C. Half of the initial volume
D. One quarter of the initial volume
E. The same as the initial volume

7.2. By what ratio does the volume of a gas increase when its temperature is increased from 9.00 °C to 27.0 °C at constant pressure?

- **Avogadro's law:** the volume and number of moles of a gas are directly proportional at constant pressure and temperature.

  Functional and linear form: $V = kn$

  Multi-point form: $\frac{V_1}{n_1} = \frac{V_2}{n_2}$

- Avogadro's law is relevant to gas-phase reactions occurring at constant temperature and pressure (for example, inside a balloon in a thermostatic room).

- When 1.00 L N_2 is combined with 3.00 L H_2…

$$N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$$

The Ideal Gas Law. Combining the results above leads to the ideal gas law or ideal gas equation of state, an equation relating *P*, *V*, *n*, and *T*. While the equation is useful for relating these variables, it is also extremely deep, as it follows from a model of the ideal gas that we'll dig into in the second half of the chapter and it includes a constant *R* that shows up in a wide variety of contexts in chemistry (most of which you'll explore in CHEM 1212K).

- The previously described **empirical gas laws** can be combined to create a universal equation of state for ideal gases known as the **ideal gas law**.

$$PV = nRT$$

- The **ideal gas constant (*R*)** has units of energy (!) per mole per Kelvin. The energy units can vary; the units of *P* and *V* should match the energy unit in *R* when we carry out calculations.

$$\begin{aligned} R &= \frac{PV}{nT} \\ &= 0.08206\ \frac{\mathrm{L \cdot atm}}{\mathrm{K \cdot mol}} \\ &= 8.314\ \frac{\mathrm{J}}{\mathrm{K \cdot mol}} \end{aligned}$$

- At constant temperature, the molarity of an ideal gas and its pressure are linearly related and *conceptually identical!* For a gas *i*,

$$[i] = \frac{n_i}{V_i} = P_i\left(\frac{1}{RT}\right)$$

Partial Pressure and Dalton's Law. For reasons that will become clear in the second half of this chapter, ideal gases in a mixture behave as if they're in the container all by themselves—each component doesn't "notice" the others. Put another way, the partial pressure of a gas in a mixture is unaffected by the presence of the other gases and the total pressure of the mixture is just the sum of the partial pressures. This idea is known as Dalton's law.

- **Partial pressure (P_i):** the pressure due to gas *i* in a mixture of gases. At constant temperature, this is proportional to the molarity [*i*] (see above).

- **Mole fraction (x_i):** moles of gas *i* divided by the total moles of gas in a mixture of gases.

$$x_i = \frac{n_i}{\sum_i n_i}$$

- **Dalton's law of partial pressures** has two aspects.

    - The total pressure of a mixture of gases is equal to the sum of the partial pressures of the constituent gases.

$$P_{tot} = P_1 + P_2 + ... + P_N = \sum_i^N P_i$$

    - Given the total pressure of a mixture of gases P_{tot} and the mole fractions x_i of the constituents, the partial pressure of each gas is the product of the total pressure and the mole fraction.

$$P_i = P_{tot} \cdot x_i = P_{tot} \left(\frac{n_i}{\sum_i n_i} \right)$$

- Keep it simple. The partial pressure of gas *i* in a mixture is unaffected by the other gases.[1]

[1] Part of your brain knows that this can't possibly be true in reality—and Dalton's law is indeed an oversimplification of the real world, especially at high total pressures. We'll explain this in class.

Example. Consider a gas cylinder containing 1.0 mol A, 0.4 mol B, and 0.6 mol C.
If the total pressure in the cylinder is 5.0 atm, what are the partial pressures of each gas?

7.3. Container A contains a gas under 3.24 atm of pressure. Container B contains a gas under 2.82 atm of pressure. Container C contains a gas under 1.21 atm of pressure. If all these gases are put into Container D, then what is the pressure in Container D in atm?

Container A = 1.23 L
Container B = 0.93 L
Container C = 1.42 L
Container D = 1.51 L

Stoichiometry of Gaseous Reactions. The ideal gas law indicates that if we know *P*, *V*, and *T* for a gas, we know *n*. Here, we're going to incorporate the ideal gas law into our existing "stoichiometry toolbox." The law takes us from the "laboratory world" of pressure, volume, and temperature measurements into "mole world."

- The ideal gas law is a useful addition to our stoichiometry toolbox, enabling pressure or volume to "stand in" for number of moles.

$$n = \frac{PV}{RT}$$

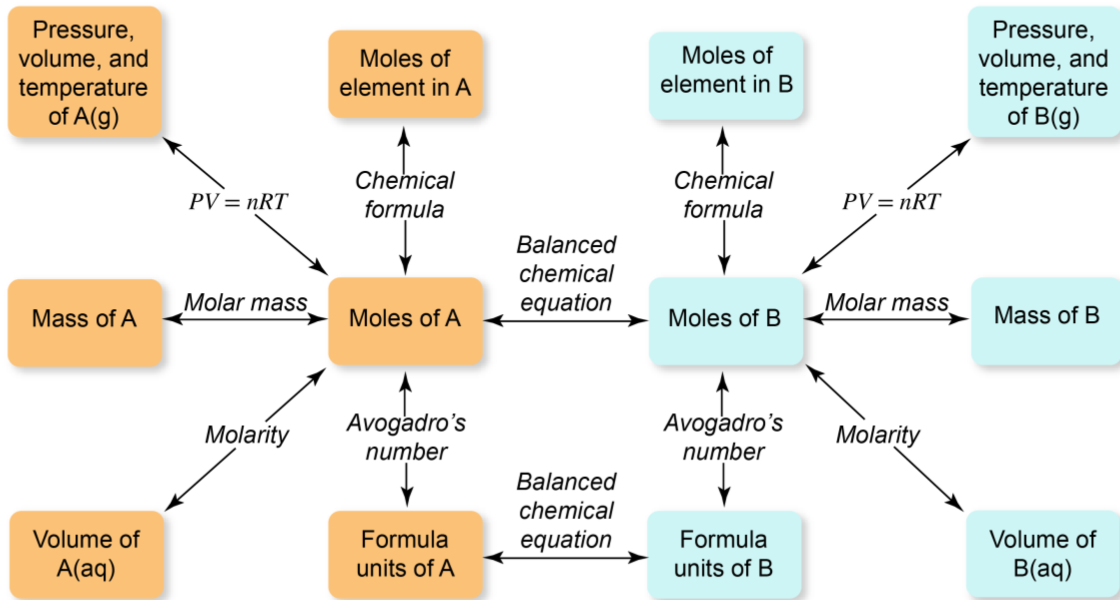


Figure 7.20. A stoichiometry toolbox for reactions involving gases.

7.4. A 5.43 g sample of an elemental diatomic gas occupies 4.38 L at 25 °C and 0.950 atm. What is the identity of the gas? Enter the molecular formula of the gas.

7.B Gases II

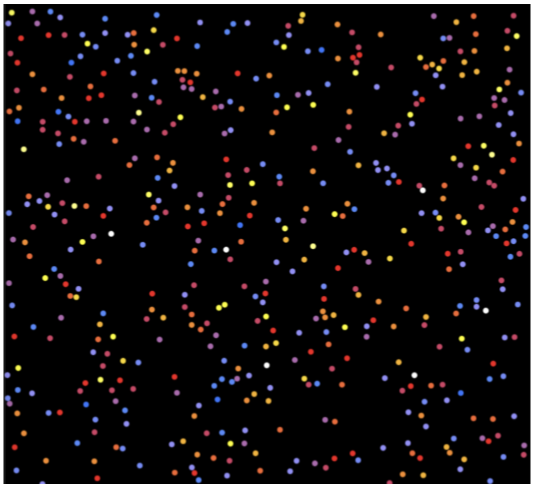
7.5. N_2O_5 is an unstable gas that decomposes according to the following balanced equation.

$$2\ N_2O_5(g) \rightarrow 4\ NO_2\ (g) + O_2(g)$$

A sample of N_2O_5 is placed into a 5.00-L container at 298.0 K. The initial pressure inside the container is 0.400 atm. What is the $P(NO_2)$ after the sample of N_2O_5 decomposes if the temperature after reaction is 310.2 K?

The Kinetic Molecular Theory of Gases. The ideal gas law follows from a submicroscopic model of the ideal gas that is used in theories of solutions, chemical kinetics, equilibrium, electrochemistry, and other areas. The model is called the kinetic molecular theory of gases; its wide use provides our motivation for learning it now.

- The ideal gas law follows from a model of the ideal gas based on the **kinetic theory of gases**.



https://falstad.com/gas

- The kinetic theory includes five key tenets or postulates.

    1. Gas particles take up no volume (i.e., particles are infinitely small points).
    Every particle can access the full volume *V* of the container.

    2. No forces exist between particles except on collision.
    Thus, all the energy of an ideal gas is kinetic energy.

    3. Collisions are perfectly elastic. No energy is dissipated or "internalized" during collisions. Collisions do not change the total kinetic energy.

    4. Particles are in constant random motion based on Newton's laws of motion.

    5. Absolute temperature (Kelvin) is proportional to the average kinetic energy of the gas particles. This is the definition of the Kelvin scale.

- Gas pressure is the result of collisions of gas particles with the container walls.

    - At lower volume and higher temperature, collisions of gas particles with the walls become more frequent and pressure rises.

- Dalton's law is readily explained using the kinetic theory.

    - Gas particles take up no volume (tenet 1) so each gas in a mixture has access to the full volume of the container…and the partial pressure that implies.

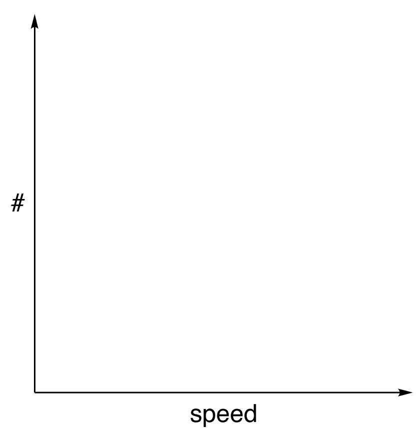
The Maxwell Distribution. Exploring the simulation of an ideal gas above quickly shows us that the distribution of speeds in an ideal gas at equilibrium is not uniform—different particles move at different speeds (or velocities *v*). Remarkably however, the distribution of speeds follows a consistent mathematical pattern first worked out by Maxwell. We will focus mostly on how the Maxwell distribution of particle speeds in an ideal gas is affected by temperature and molar mass.

- The distribution of particle speeds in an ideal gas is called the **Maxwell distribution**. It reveals the probability of encountering a particle with each possible speed value in an ideal gas at equilibrium.[2]

- The **most probable speed (v_p)** is located at the peak of the distribution.

- The **root mean square speed (v_{rms})** is the square root of the average of squared particle speeds. The average kinetic energy of gas particles can be written in terms of v_{rms}—this is why it's useful.

$$v_{rms} = \sqrt{\frac{\sum_i^N v_i^2}{N}} \qquad\qquad \overline{KE} = \frac{1}{N}\sum_i^N \frac{mv_i^2}{2} = \frac{mv_{rms}^2}{2}$$

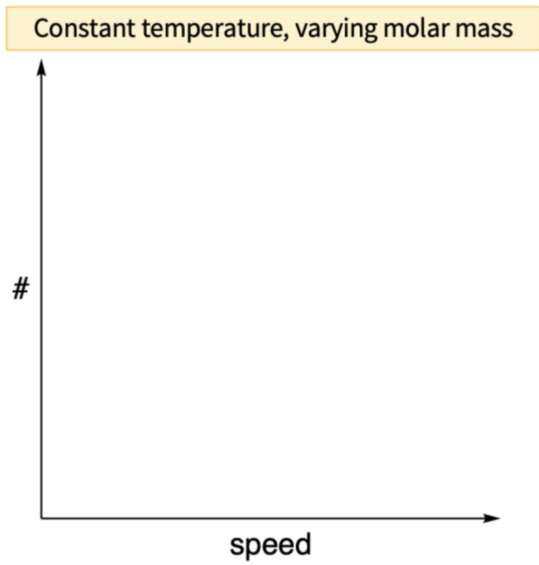


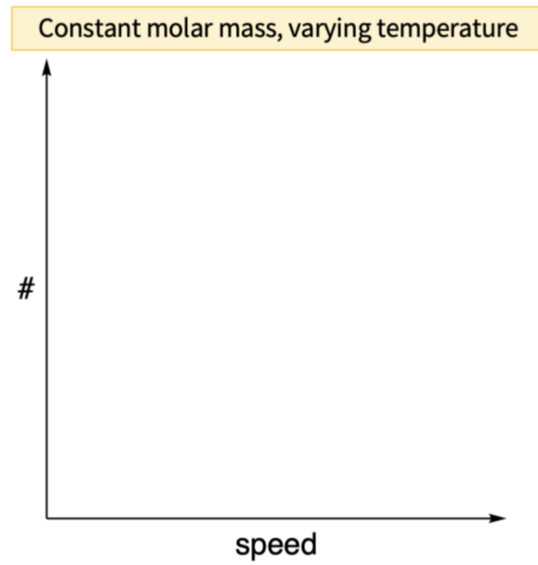
- The analytical form of the distribution is the following. Note the appearance of particle mass *m* and temperature *T* and the essential role of particle speed in $v^2 \exp(-v^2)$. The essence of the **Boltzmann constant k_B** will soon become clear…

$$f(v) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi v^2\, e^{-\frac{mv^2}{2k_B T}}$$

[2] It helps to think of these distribution graphs as "continuous histograms." At each value *v* along the horizontal axis, the height of the curve indicates the number of particles with velocity between *v* and *v* + *dv* (an infinitesimal "sliver" of the velocity axis).

- Most probable speed and root mean square speed depend on temperature and molar mass.





- Gas particles generally travel only a very short distance before colliding with either another particle or the container wall and changing direction.

- **Mean free path** is the average distance traveled by a gas particle between collisions.

7.6. Which of the following statements is/are *true* of ideal gases?

A. When molecules of ideal gas collide, energy is exchanged between them.
B. Molecules of F_2 gas and Cl_2 gas have the same kinetic energy at 875 K.
C. Particles of gas are more attracted to one another at 1000 K than at 300 K because the particles at 1000 K have greater kinetic energy.
D. Kinetic-molecular theory adequately explains why increasing the number of moles of gas in a sample increases pressure.
E. Increasing pressure causes gases to behave more ideally because it makes the volume of gas particles appreciable with respect to the volume of the container.

Diffusion and Effusion. The Maxwell distribution governs the speeds of particles in an ideal gas. From it, we can derive several important ideas related to processes in which gas particles are moving through space. Specifically, the Maxwell distribution gives us insight into diffusion, the bulk movement of a gas through space, and effusion, the escape of gas particles through a tiny hole into vacuum.

- **Diffusion** is the process by which gas particles spread out in response to a gradient in concentration.
    - Diffusion is faster at greater root mean square speed v_{rms}. Faster-moving gas particles result in faster diffusion of the bulk gas (a "duh!" thing for us, but not for Maxwell).
    - The molar heat capacity of an ideal gas is (3/2)R. Because all the energy of the gas is kinetic energy, we can relate temperature and molar mass M to the root mean square speed as follows.

$$\overline{KE} = \frac{3}{2}RT = \frac{Mv_{rms}^2}{2}$$

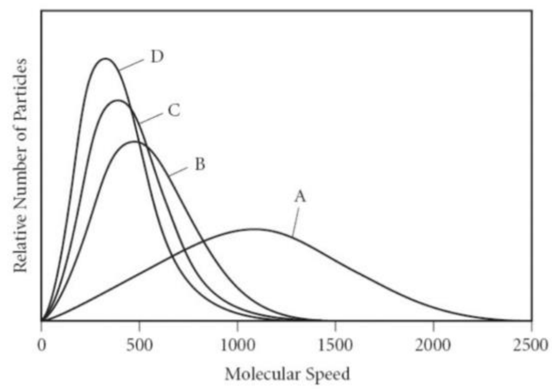
$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

    - The unsurprising conclusion is that diffusion is faster at higher temperatures and smaller molecular weights.

- **Effusion** is the one-way escape of a gas from a container through a tiny hole into vacuum.
    - **Graham's law of effusion** states that effusion rate r is inversely proportional to the square root of molar mass M.

$$r \propto \sqrt{\frac{1}{M}} \qquad\qquad \frac{r_2}{r_1} = \sqrt{\frac{M_1}{M_2}}$$

    - Compare to our results for diffusion above!



7.7. The graph above represents four different gases at the same temperature. Which of the following statements are *true*?

A. Molecules of the gas represented by curve A have a greater distribution of speeds due to the gas having a larger molar mass.

B. Molecules of the gas represented by curve C have a smaller molar mass than the molecules represent by gas B.

C. All the curves actually represent the same compound at the same temperature, but have different root mean square speeds.

D. Molecules of the gas represented by curve D have the smallest root mean square speed and would diffuse the fastest.

E. The molecules of gas B will effuse faster than the molecules of gas C.

7.8. A mixture of O_2(g) and SO_2(g) in a gas cylinder is allowed to effuse through a tiny orifice. Suppose that in some time frame, 1.00 mol of O_2(g) have effused out of the cylinder. How many moles of SO_2(g) have effused out in that same time frame?

Real Gases. The kinetic theory of gases is an idealization of gas particles that breaks down in many "real-world" situations. Real gases don't conform to the ideal gas equation of state at all temperatures and pressures. Here, we'll learn a metric for assessing "how ideal" a gas is under given conditions, the submicroscopic reasons why the ideal gas model breaks down, and an improved equation of state that works under a broader range of *T* and *P* conditions.

- The tenets of the kinetic molecular theory don't always hold…
  - Gas particles *do* take up volume!
  - Gas particles *do* interact with each other.
    There are forces between particles!
- Particularly at high pressures and low temperatures,
  gases tend to violate the ideal gas law (behave nonideally).
- **Compressibility factor *Z* (*PV*/*RT*)** provides a measure of ideality.
  $Z = 1$ corresponds to ideal behavior.

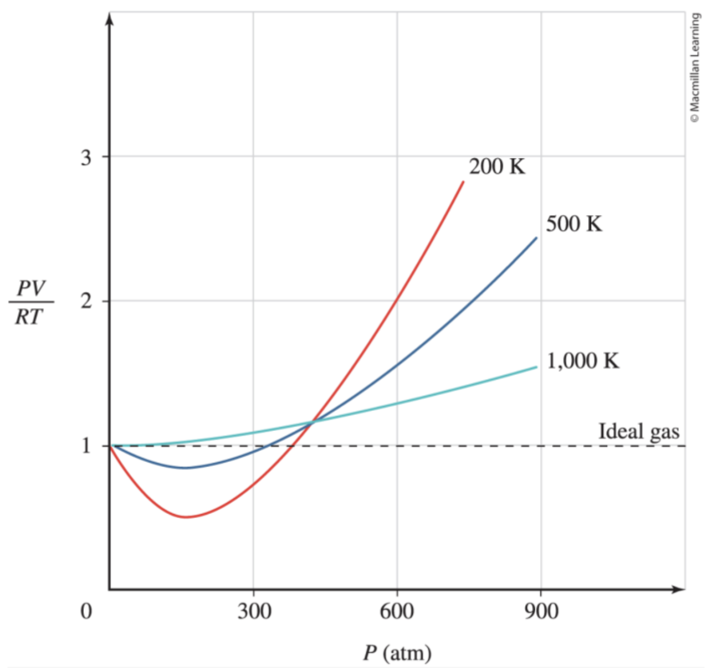


Figure 7.26. Compressibility factor *PV*/*RT* as a function of pressure and temperature for N_2 gas.

- Particles of a real gas are attracted to one another, although these attractions tend to be weak because of the long distances between the particles and their rapid speed.
    - At higher temperatures…
    - At lower temperatures…

- The **van der Waals** equation of state is a modification of the ideal gas law that incorporates parameters related to interparticle forces (*a*) and the volume taken up by gas particles (*b*).

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$$

    - The constant *a* is a measure of intermolecular force strength with units of energy·mol^{-1}·M^{-1}.
    - The constant *b* is a measure of the volume taken up by gas particles with units of volume·mol^{-1}.

Chapter Summary. Some of the amazing things we can now do with gases…

1. Define and interpret pressure; express pressure in a variety of units.

2. Carry out calculations based on the ideal gas equation of state and the empirical gas laws.

3. Carry out stoichiometric calculations for gaseous reactions.

4. Articulate and apply the tenets of the kinetic theory of gases.

5. Use the kinetic theory to explain the behavior of gases.

6. Describe the distribution of particle speeds in an ideal gas at equilibrium; predict how changes in molar mass and/or temperature will affect that distribution.

7. Relate rates of diffusion and effusion to root mean square speed, temperature, and molar mass.

8. Explain how real gases violate the tenets of the kinetic theory; recognize conditions under which gases behave less ideally.

9. Apply the van der Waals equation of state to calculate variables related to real gases.